

Refer to [Section 5.3.14: I/O port characteristics](#): I/O port characteristics for more details on the input/output alternate function characteristics (CK, SDO, SDI, WS).

Table 68. I2S characteristics

Evaluated by characterization – Not tested in production.

Symbol	Parameter	Conditions	Min	Max	Unit
f_{MCK}	I2S main clock output	-	-	50	MHz
f_{CK}	I2S clock frequency	Master Tx or RX, slave Rx	-	50	MHz
		Slave Tx	-	19	
$t_{V(WS)}$	WS valid time	Master mode	-	3	ns
$t_{H(WS)}$	WS hold time	Master mode	1	-	
$t_{SU(WS)}$	WS setup time	Slave mode	2	-	
$t_{H(WS)}$	WS hold time	Slave mode	0.5	-	
$t_{SU(SD_MR)}$	Data input setup time	Master receiver	2.5	-	ns
$t_{SU(SD_SR)}$	Data input setup time	Slave receiver	1	-	
$t_{H(SD_MR)}$	Data input hold time	Master receiver	1.5	-	
$t_{H(SD_SR)}$		Slave receiver	2.5	-	
$t_{V(SD_ST)}$	Data output valid time	Slave transmitter (after enable edge)	-	15.5	
$t_{H(SD_ST)}$	Data output hold time	Slave transmitter (after enable edge)	8	-	
$t_{V(SD_MT)}$	Data output valid time	Master transmitter (after enable edge)	-	2	
$t_{H(SD_MT)}$	Data output hold time	Master transmitter (after enable edge)	0.5	-	

5.3.27 USB_FS characteristics

Table 69. USB_FS characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_{DDUSB}	USB transceiver operating supply voltage	-	3.0 ⁽¹⁾	-	3.6	V
R_{PUI}	Embedded USB_DP pullup value during idle	-	900	-	1575	Ω
R_{PUR}	Embedded USB_DP pullup value during reception	-	1425	-	3090	
Z_{DRV}	Output driver impedance ⁽²⁾	High and low driver	28	36	44	

1. USB functionality is ensured down to 2.7 V, but some USB electrical characteristics are degraded in 2.7 to 3.0 V range.
2. No external termination series resistors are required on USB_DP (D+) and USB_DM (D-). The matching impedance is already included in the embedded driver.

5.3.28 JTAG/SWD interface characteristics

Unless otherwise specified, the parameters given in [Table 70](#) and [Table 71](#) are derived from tests performed under the ambient temperature, f_{HCLKx} frequency and V_{DD} supply voltage conditions summarized in [Table 18](#), with the following configuration:

- Output speed set to $OSPEEDRy[1:0] = 10$
- Capacitive load $C_L = 30$ pF
- Measurement points done at $0.5 \times V_{DD}$ level

Refer to [Table 48. I/O static characteristics](#) for more details on the input/output characteristics.